7. Which of the following fluro compounds is most likely to behave as a Lewis base?

(NEET – 2016)

a) BF_3 b) PF_3 c) CF_4 d) SiF_4

8. Which of these is not likely to act as Lewis base?

a) BF_3 b) PF_3 c) CO d) F^-

9. The aqueous solutions of sodium formate, anilinium chloride and potassium cyanide are respectively

a) acidic, acidic, basic b) basic, acidic, basic

c) basic, neutral, basic d) none of these

10. The percentage of pyridine (C_5H_5N) that forms pyridinium ion (C_5H_5NH) in a 0.10M aqueous pyridine solution (K_b for $C_5H_5N = 1.7\times10^{-9}$) is

a) 0.006% b) 0.013% c) 0.77% d) 1.6%

11. Equal volumes of three acid solutions of pH 1,2 and 3 are mixed in a vessel. What will be the H^+ ion concentration in the mixture?

a) 3.7×10^{-2} b) 10^{-6} c) 0.111 d) none of these

12. The solubility of AgCl (s) with solubility product 1.6×10^{-10} in 0.1M NaCl solution would be

a) 1.26×10^{-5}M b) 1.6×10^{-9}M c) 1.6×10^{-11}M d) Zero

13. If the solubility product of lead iodide is 3.2×10^{-8}, its solubility will be

a) 2×10^{-3}M b) 4×10^{-4}M c) 1.6×10^{-5}M d) 1.8×10^{-5}M

14. MY and NY_3, are insoluble salts and have the same K_{sp} values of 6.2×10^{-13} at room temperature. Which statement would be true with regard to MY and NY_3?

a) The salts MY and NY_3 are more soluble in 0.5M KY than in pure water

b) The addition of the salt of KY to the suspension of MY and NY_3 will have no effect on their solubility's

c) The molar solubilities of MY and NY_3 in water are identical

d) The molar solubility of MY in water is less than that of NY_3

15. What is the pH of the resulting solution when equal volumes of 0.1M NaOH and 0.01M HCl are mixed?

a) 2.0 b) 3 c) 7.0 d) 12.65

16. The dissociation constant of a weak acid is 1×10^{-3}. In order to prepare a buffer solution with a pH = 4, the $\frac{[Acid]}{[Salt]}$ ratio should be

a) 4:3 b) 3:4 c) 10:1 d) 1:10

XII U8-Ionic equilibrium.indd 29 11-01-2021 12:59:25